

A Reflective Journey: Navigating Your Cumulative Experience at Iowa State University

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Foreword

Through my experiences in coursework and industry, I have learned that analog integrated circuits do not immediately emerge fully developed and proven. The design process begins with architectural choices informed by specifications. From there, early design decisions establish DC operating points, followed by the design and optimization of gain and feedback. Once an ideal design has been established, the nonidealities of the circuit, such as parasitics and nonlinear transistor effects, must be investigated and remediated. Full characterization then reveals, through extensive testing, the circuit's behavior in a myriad of operating conditions. Potential issues are then addressed through iterative redesign, before the design is ready for tapeout. This design process is a recursive loop of learning, deciding, trying, failing, and adjusting.

I've come to see my undergraduate education as a similar process. I enrolled at Iowa State University with a general interest in electrical engineering, but lacking a clear goal. Through coursework, extracurricular involvement, internships, projects, and peers, my understanding of engineering and my goals have repeatedly changed. During this process, my goals were eventually defined, redefined, and refined, allowing me to proceed with a concrete goal in mind.

The following sections trace my development as a student at ISU using stages of the analog circuit design process as a framework, from the **DC Operating Point** to the final **Tapeout**.

I – DC Operating Point

Setting the DC operating point of an analog circuit determines many facets of its subsequent behavior and character. It establishes the conditions of the circuit, influencing quantities such as gain, bandwidth, and linearity. When I first moved to Ames to start my degree, I had not established my proverbial operating point for myself: I questioned, “What would I like to spend a vast portion of my life doing?”

I knew that I enjoyed mathematics and physics, and I was quite proficient at both, even completing all my required calculus classes while still in high school. I also knew that electrical engineering seemed to provide a useful intersection of both, where I would be able to find employment related to these nascent interests. My early conception of electrical engineering was broad and somewhat abstract. I had little understanding of either the technology itself or the activities in designing and integrating it.

An operating point is not something that exists independently of the circuit(s) around it, but rather it is established by the surrounding conditions. My own operating point began to emerge through exposure: first to fundamentals, then to increasingly specialized knowledge and practice, even in industry experience.

The important growth I experienced in my education was the accumulation of diverse knowledge, which allowed me to make increasingly informed decisions about what I wanted from engineering.

When faced with my first integrated circuits class, I knew I had found a niche in the field that I felt a strong passion for. I spent my first couple of semesters just focusing on performing well on exams and getting good grades in my classes that I had not really paid proper attention to the larger implications of my coursework. Eventually, through constant exposure to new topics, I found my operating point.

II – Gain

An amplifier provides gain by transforming an oftentimes very small signal into a much larger output signal. The input signal might be nigh imperceptible, but through the magnifying lens of an amplifier, it becomes consequential and more useful. My education at Iowa State similarly amplified my limited knowledge of crucial basic principles into a broader body of technical knowledge, greatly developing skills required for complex problem-solving.

My knowledge progression was cumulative. Early courses in the program established the underlying mathematical principles of electrical engineering topics, such as circuit analysis and digital logic. As I advanced through the courses, these previously introduced topics began to intersect further. Small-signal models, while introduced in a previous class, were given better motivation and use in my integrated circuits course. Feedback theory became implementable and useful, rather than motivated by a strictly theoretical approach.

My internship experiences reinforced this relationship between theory and practice even further. I spent two summers at Collins Aerospace, working with hardware associated with GPS and vehicle navigation applications. At that time, I encountered practical requirements surrounding circuit cards and hardware validation. I learned that theoretical analysis of many schematics may not be too difficult, but modeling and anticipating their behavior on a printed circuit board is a totally different skill.

My internship in the summer of 2026 at Skyworks provided the most direct and meaningful “amplification” of my analog integrated circuit design skills. Working on a team in charge of designing power amplifier controller circuits exposed me to real transistor-level designs in actual semiconductor process technologies. The theoretical knowledge obtained in my integrated circuits class was suddenly all incredibly useful and important, encouraging the further growth of my practical skills.

III – Feedback

Feedback allows an analog circuit to compare its current behavior with desired conditions and adjust accordingly. With proper application, it provides greater stability and accuracy. Feedback, in its many forms, has been similarly crucial in my development as a student and professional.

Professors, coworkers, peers, and unexpected simulation results have all provided me with amorphous feedback, fostering the necessary ecosystem for my technical growth.

The most important feedback in my undergraduate studies has come from professors willing to engage with questions that extended beyond the coursework. During my freshman year, I was able to gain some experience in a research lab via the First-Year Honors Mentor Program. Assisting in biomedical photoacoustic tomography research with Dr. Manojit Pramanik required me to understand an existing system with practical limitations and develop hardware that could improve it. While still in my first year of study, I was able to get exposure and experience with designing a printed circuit board, before I really had learned much about these topics in class. It was an eye-opening experience to work on developing technology that could later be used in a medical setting.

Industry provided another form of feedback to me. During my internships, I worked alongside engineers whose designs had to be shaped by a wide variety of stringent guidelines. At Skyworks, leading design reviews and having discussions with coworkers both forced me to learn the necessary communicative skills to prove my knowledge and skills and provided me with countless avenues for professional feedback.

Design reviews are full of competent and experienced designers, all with their eyes on my own designs, with everyone interested in improving performance and area, along with making certain that the design will perform as needed. If a circuit displayed some sort of unexpected or undefined behavior, the expectation was for me to have thoroughly investigated and characterized the issue, so that the other professionals in the room could give feedback on the design.

Peers have been similarly important as a source of feedback. Engineering education is by no means an individual pursuit, as many students working together is hugely beneficial for all involved. If one student is confused, the variety of other perspectives that are available from other students is almost certain to provide them with some explanation that they find intuitive.

IV – Parasitics

An analog circuit is gestated as an abstraction: a schematic of lumped components co-existing in a netlist, an idealized representation of the circuit. Once the circuit is physically shaped, however, the abstraction garners nonidealities previously absent from the design. Parasitic capacitances, interconnect resistances, coupling, and other effects were not yet apparent in the schematic. My education and exposure to curricula followed a similar path.

Earlier in the degree program, classes introduced fundamental concepts in a vacuum. This was necessary to first acquire a solid understanding before adding too many moving parts. As I advanced through my classes, the focus shifted from pure theoretical understanding towards a more practical approach to analysis.

My industry experience in integrated circuit design at Skyworks made this distinction incredibly apparent. Circuit schematics suggest that a circuit must behave a certain way, but its physical design/layout can often manipulate the behavior greatly, affecting timing, stability, current leakage, etc. During my time at Skyworks, I encountered these issues frequently in simulations that accounted for parasitic extractions, showing that transient behavior often degrades with the additional nonidealities introduced by interconnect and parasitic capacitances.

The same progression occurred in a more pragmatic manner, removed from the strictly technical/theoretical situations. In coursework, certain problems are simplified to more accurately isolate the material being taught at that moment. In electromagnetics, you assume conductors are lossless and perfectly conductive. In circuit analysis, you assume steady-state, or you assume that capacitors have initial conditions. In project work, these nominal representations of components and situations need to dissolve or evolve in accordance with real constraints. Sometimes components or devices are too expensive, and costs need to be lowered elsewhere, or maybe manufacturing time will take too long, so adjustments need to be made. These previously unseen limitations have become increasingly apparent to me through my experiences at Iowa State and my internships.

These constraints also extend beyond the immediate circuit or project. Semiconductor design exists within a global manufacturing ecosystem. Decisions about process technology, power consumption, and material use have consequences extending well beyond the individual design. I have become more aware that an engineering solution must be evaluated for functionality, along with whether it can be produced and used responsibly at scale, as all designs, big or small, can have lasting effects on the global economy and the environment.

V – Nonlinearity

Analog circuits are built from nonlinear devices, regardless of what the simplified device models used in hand calculations might suggest. Those approximations are in fact useful precisely because they neglect the nonlinearities, making them easier to learn and understand. Past their pedagogical utility, however, those simplified models have limits. Sometimes the complication is not something omitted from the schematic, but an assumption within the underlying model that no longer holds.

My education followed a similar progression, but this is not the only domain in which my preconceived models proved insufficient. When I came to Iowa State, I had assumptions about what engineering would entail, what I would be good at, where those skills might lead me, and how my life would change and evolve. Those assumptions served a purpose, but through experience in both coursework and industry, my initial “models” could not adequately account for what I encountered.

If I were starting my undergraduate studies again, I would spend less time trying to determine what kind of specialty I’d like to pursue, and I would spend more time trying to pursue those areas. I

would seek more opportunities for difficulties and experience, earlier in my education. The real growth for me came from those undefined moments where I was not sure how to proceed.

The analogous “model” I came into my education with was the idea that career trajectory is determined solely by what classes one takes, or that those decisions are more permanent than they truly are. Through my own experience, moving from classes focused on digital signal processing and communication protocols to RF circuits and integrated circuit design, along with my industry change from aerospace printed circuit board design to analog integrated circuits for mobile phones, I learned that this metaphorical “model” was far too simple and did not account for the many other factors, seen and unseen, that go into finding one’s career.

This has changed the way I approach unfamiliar opportunities. I no longer seek the experience to fit neatly into my existing understanding of what I should be doing, and I instead allow it to better define my existing understanding. I can see that my progression from aerospace hardware to RF and integrated circuit design was not a sequence of random deviations from some original plan, and it was rather a process by which the plan itself was made more accurate.

VI – Characterization

Designing a circuit is defining what it is intended to do; characterization is the process of observing how it truly behaves. By examining its operation across a variety of conditions, variations, and imperfections, engineers can identify whether a design is robust. Through my undergraduate education, I have learned to extensively validate ideas before relying on them, trusting my understanding of a system only when it survives reality.

My internships provided repeated opportunities to really develop and harden this habit. At Collins Aerospace, failure analysis of hardware required using test information to determine issues with the circuit boards. In the manufacturing world, data and automated testing provided additional information about system performance that went far beyond what the schematic alone could provide.

At Skyworks, characterization became more directly tied to transistor-level circuit design. Simulating over process variations, voltage changes, and temperature deviations allowed me to examine changes in circuit performance in different environments. I also used Monte Carlo analysis, which repeatedly simulates the circuit but with additional modeling based on expected manufacturing distributions. Rather than showing behavior in ideal conditions, this provides an estimate of how consistent this design will be in many manufactured instances.

A circuit that performs well at normal operating conditions is maybe useful, but the robustness is what proves the design. Thorough characterization therefore required that I evaluate circuits in realistic conditions and design for such conditions. This process reinforced the importance of

grounded engineering decisions, informed by plenty of evidence. Before accepting a conclusion, I increasingly ask how much evidence supports it and what conditions may invalidate it.

VII – Iteration

It must now be very evident that very few analog circuits are perfect on the first iteration. A design is simulated, characterized, revised, tested, and redesigned repeatedly, with each version incorporating information learned from the last. Engineering therefore not only requires solving problems once, but repeatedly doing so while remaining willing to alter a design in response to new information. My education at Iowa State has undergone this cycle repeatedly, with each experience informing the next and shaping how I approach unfamiliar problems and situations.

This experience has also shaped how I learn and facilitate my own education. Earlier in my education, my primary strategy was knowing what I needed to know for exams to keep my GPA up. That approach worked for school but was bounded by the ideal nature of exam questions and was less applicable to real problem-solving.

Now, when I encounter something that I do not understand, I am a lot more willing to construct my own curriculum or learning process around it. I often refer to textbooks, research papers, simulation results, course material, and conversations with peers. My analog IC design studies continue to extend beyond coursework, as I venture into the texts of Razavi, along with Gray, Hurst, Lewis, and Meyer (P. R. Gray, P. J. Hurst, S. H. Lewis and R. G. Meyer, “Analysis and Design of Analog Integrated Circuits,” 4th Edition, John Wiley and Sons Inc., New York, 2001.). Reading these textbooks on my own time allows a more relaxed environment for me to learn the information I’m so passionate about.

Along with these textbooks, I found myself referring to lots of technical documentation and research papers while at Skyworks. It was not uncommon to find myself referring to their in-office library to find information about circuit topologies or referring to papers online about different types of op-amp compensation feedback networks.

My education process has become increasingly self-directed. As I learn my own limits and strengths, I am able to guide myself towards useful resources that enhance my coursework education. This iterative approach has shaped my ability to learn extensively about relevant topics.

The willingness to find and create my own learning opportunities is one of the more important skills I will take from my time at Iowa State. The technologies, software, and hardware that I work with will likely change drastically over time, but this iterative approach to learning will allow me to continue to learn for years to come.

VIII – Tapeout

Tapeout is the culmination of the integrated circuit design process: a design has graduated its abstractions and has been realized in its physical form. It is a milestone of completion, yes, but only the beginning of the next step in a larger cycle. The fabricated circuit will be tested and potentially redesigned, then ultimately serve a new purpose.

As I approach the end of my undergraduate education, I view it similarly: as a completion through which one phase of my life will transform into the next. My development as an engineer is still only beginning, as I undergo my own metamorphosis into a professional.

My immediate goal is to continue developing in analog and mixed-signal integrated circuit design. Immediately upon graduation, I have already arranged that I will return to Skyworks as a graduate intern, until I attend graduate school in fall 2027. I am in the process of applications, and I am not yet sure where that will be. Graduate studies will allow me to deepen my technical foundation that I have germinated at Iowa State. I expect that this process will begin another period of realizing how much I do not know and must learn, but I anticipate that this will be less uncomfortable after having experienced that realization many times as an undergraduate. I have become more comfortable treating that realization as cause for further exploration rather than evidence that I am unprepared and incompetent.

There are still many technical areas in which I plan to grow. I want to acquire a deeper understanding of analog circuit architecture, noise, and RF and high-frequency integrated circuits. I also want to keep developing my communicative skills, so I can fluently articulate my ideas.

More broadly, I want to retain the habits that I have developed and that have served me best during undergraduate studies at Iowa State. I intend to seek difficult problems, ask for feedback, and revise when reality challenges my perceptions.

As previously mentioned, a tapeout marks the point at which an engineer releases a design into the physical world, where its behavior is no longer solely determined by nominal intention. I'm at a similar point as I finish my bachelor's degree in EE at Iowa State. I will leave with considerably more knowledge than I came with, but also with a better idea of how much I have left to learn. Tapeout is a milestone, not an endpoint, and I intend to approach my own development in much the same way.